

Intermediate Macro

Chapter 6: Financial Markets II

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6.1. N. = 1, LV = 1, D = 1, L = 1.

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THE FIRST AND SECOND VOLUMES

[illegible]

6.4.1. **U.S. and U.K. Data**



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6.1 Nominal Versus Real Interest Rates

- In January 1980, the one-year US T-bill rate was 10.9%. In January 2006, the rate was only 4.2%. It was clearly much cheaper to borrow in 2006 than it was in 1981.

Nominal and Real Interest Rates in the United States

6.1 Nominal Versus Real Interest Rates

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6.1 Nominal Versus Real Interest Rates

- In January 1980, the one-year US T-bill rate was 10.9%. In January 2006, the rate was only 4.2%. It was clearly much cheaper to borrow in 2006 than it was in 1981. Or, was it?
- When we borrow or lend, what we really want to know is **how many goods**—not how many dollars—we will have to give up or get in the future in exchange for the goods we get or give up today.

Definition of i_t and r_t

6.1 Nominal Versus Real Interest Rates

- Interest rates expressed in terms of dollars are called **nominal interest rates**. If the nominal interest rate for year t is i_t , borrowing 1 dollar this year requires you to pay $1 + i_t$ dollars next year.
- Interest rates expressed in terms of a basket of goods are called **real interest rates**. If we denote the real interest rate for year t by r_t , borrowing the equivalent of one basket of goods this year requires you to pay the equivalent of $1 + r_t$ baskets of goods next year.

6.1 Nominal Versus Real Interest Rates



Relation Between i_t and r_t

6.1 Nominal Versus Real Interest Rates

- Putting together what we see in the previous discussion, it follows that the one-year real interest rate r_t , is given by:

$$1 + r_t = (1 + i_t) \frac{P_t}{P_{t+1}^e} \quad (6.1)$$

- Denoting expected inflation between t and $t + 1$ by π_{t+1}^e , we have

$$\pi_{t+1}^e = \frac{P_{t+1}^e - P_t}{P_t} \quad (6.2)$$

- Using equation (6.2), rewrite $\frac{P_t}{P_{t+1}^e}$ in equation (6.1). Replace in equation (6.1) to get

$$1 + r_t = \frac{1 + i_t}{1 + \pi_{t+1}^e} \quad (6.3)$$

An Approximated Expression

6.1 Nominal Versus Real Interest Rates

- Equation (6.3) gives us the exact relation of the real interest rate to the nominal interest rate and expected inflation. However, when both the nominal interest rate and expected inflation are not too large—say, less than 10% per year—a close approximation to this equation is given by

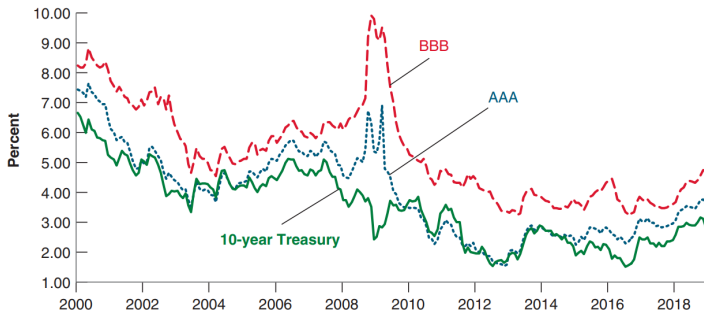
$$r_t \approx i_t - \pi_{t+1}^e \quad (6.4)$$

- This equation has several implications:
 - When the expected inflation rate is zero, the nominal rate and the real rate are equal.
 - Since inflation is typically positive, the real rate is usually lower than the nominal rate.
 - For a given nominal interest rate**, higher inflation will lead to a lower real interest rate.

Various Types of Bonds

6.2 Risk and Risk Premia

- Until now we have assumed only one type of bond exists. However, bonds differ based on several factors including maturity and default risk.



Yields on 10-Year US Government Treasury, AAA, and BBB Corporate Bonds, since 2000

Two Factors Determining Risk Premia

6.2 Risk and Risk Premia

- Bond investors will demand a **risk premium** to hold bonds with higher levels of default risk.
- Two factors determine the risk premium:
 - Probability of default. Higher probabilities of default require higher interest rates to entice bond buyers.

Let i be the nominal interest rate on a risk-free bond, and $i + x$ be the nominal interest rate on a risky bond with the probability of defaulting p . Call x the risk premium. For risk neutral investors, the following relation must hold:

$$1 + i = (1 - p)(1 + i + x) + p \cdot 0$$

Rearranging terms gives:

$$x = \frac{(1 + i)p}{1 - p}$$

- Degree of risk aversion of the bond holders.

Understanding How Banks Operate

6.3 The Role of Financial Intermediaries

- Until now we have looked at **direct finance**, or borrowing and lending between two parties.
- Most financial transactions occur through financial intermediaries like banks and credit unions. These intermediaries play an important role in the economy by matching borrowers and lenders.
- However, as illustrated by the financial crisis, they can run into trouble.

The Choice of Leverage

6.3 The Role of Financial Intermediaries

Bank Balance Sheet

Assets 100	Liabilities 80
	Capital 20

- The **capital ratio** of a bank is defined as the ratio of its capital to its assets. In the above example, the capital ratio is $\frac{20}{100} = 20\%$. The **leverage ratio** of a bank is defined as the ratio of assets to capital, so as the inverse of the capital ratio. In this case, $\frac{100}{20} = 5$.
- In thinking about what leverage ratio it should choose, the bank faces a trade-off problem. A higher leverage ratio implies a higher expected profit rate. But it also implies a higher risk of bankruptcy.

Example

6.3 The Role of Financial Intermediaries

- Suppose the expected rate of return on assets is 5% and the expected rate of return on liabilities is 4%. Then, the expected profit of the bank is equal to $100 \times 5\% - 80 \times 4\% = 1.8\%$.
- Given that the owners of the bank have put in 20 of their own funds, the expected profit per unit of capital is equal to $1.8\%/20 = 9\%$.
- Now suppose the owners of the bank decided instead to put in only 10 of their own funds and borrowed 90.
- By increasing its leverage, and decreasing its own funds, the bank would increase its expected profit per unit of capital.
- Would it be okay for the bank to choose a higher leverage ratio? Why shouldn't?

Leverage and Lending

6.3 The Role of Financial Intermediaries

- During the crisis, many banks opted for a higher leverage ratio and thus more risk making it more likely that they would go bankrupt.
- As the bank's assets declined in value, many banks ended up in a situation where the value of their assets fell below the value of their liabilities.
- This technical **insolvency** eliminated these banks' ability to lend, even to good customers.
- Some of these bank assets were very difficult to value which resulted in the inability to sell the assets, or forced them to sell at very low prices, often referred to as **fire sale prices**.
- Investors also became worried about the banks' long-term solvency and the safety of their deposits which resulted in rapid requests to withdraw **demand deposits** and created **bank runs**.

Revisiting the $IS - LM$ Model

6.4 Extending the $IS - LM$ Model

- Now we revisit the $IS - LM$ model with the addition of inflation and risk premia embedded in market interest rates.
- Now our model looks like this:

$$IS \text{ relation : } Y = C(Y - T) + I(Y, i - \pi^e + x) + G$$

$$LM \text{ relation : } i = \bar{i}$$

- Note that LM relation remains the same. The central bank still controls the nominal interest rate.
- However, the IS relation changed in two ways:
 - ① Spending decisions depend on the real interest rate rather than the nominal interest rate.
 - ② The risk premium x captures higher perceived risk and/or risk aversion.

Representing a Graph in $r - Y$ Plane

6.4 Extending the $IS - LM$ Model

- We can now see that the interest in the LM and IS equations are different rates. The LM interest rate is the **policy rate** (nominal rate) while the IS rate is the **borrowing rate** (real rate).
- We can think of the central banks as choosing the real policy rate directly and rewrite the two equations as:

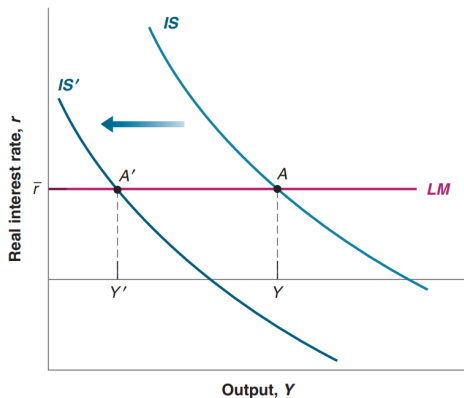
$$IS \text{ relation : } Y = C(Y - T) + I(Y, r + x) + G \quad (6.5)$$

$$LM \text{ relation : } r = \bar{r} \quad (6.6)$$

Financial Shocks and Output

6.4 Extending the $IS - LM$ Model

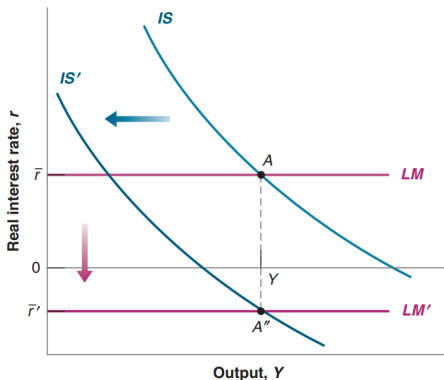
- An increase in the risk premium x leads to a shift of IS curve to the left and a decrease in equilibrium output.



Financial Shocks and Policies

6.4 Extending the $IS - LM$ Model

- If sufficiently large, a decrease in the policy rate can in principle offset the increase in the risk premium. The zero lower bound may, however, put a limit on the decrease in the real policy rate.



Housing Prices and Subprime Mortgages

6.5 From a Housing Problem to a Financial Crisis

- Falling housing prices beginning in 2006 started a process leading to the financial crisis. Prior to that point in time U.S. housing prices increased due to low interest rates, high demand and **mortgages lenders** willing to lend to risky borrowers through what is known as **subprime mortgages (subprimes)**.



Source: FRED: Case-Shiller Home Price Indices. 10-city home price index, Series SPCS10RSA

What is a Mortgage?

6.5 From a Housing Problem to a Financial Crisis



- A loan used to purchase or maintain a home, land, or other types of real estate, secured by the property itself
- The property serves as **collateral** to secure the loan.
- Mortgage applications go through a rigorous underwriting process, including credit checks and assessments of the capability of repayment of the borrower.

Terminologies

6.5 From a Housing Problem to a Financial Crisis

- **securitization**: Securitization is the procedure where an issuer designs a marketable financial instrument by merging or pooling various financial assets into one group.
- In theory, any financial asset can be securitized—that is, turned into a tradeable, fungible item of money value.
- **Collateralized Debt Obligation (CDO)**: A CDO is a complex structured finance product that is backed by a pool of loans and other assets.
- A CDO is a particular type of derivative because its value is derived from another underlying asset.
- These assets become the collateral if the loan defaults.

6.5 From a Housing Problem to a Financial Crisis



Understanding the Risky Structure of Subprime Mortgages

6.5 From a Housing Problem to a Financial Crisis

- After 2006, housing prices begin to decline. Once the decline occurred many mortgages were considered to be **underwater** (this is when the value of the mortgage exceeds the value of the house)
- It was realized that the mortgages offered were in fact much riskier than either the lender pretended or the borrower understood.
- This in turn caused many borrowers to default on their mortgages thus creating a large loss for many banks.

The Role of Financial Intermediaries

6.5 From a Housing Problem to a Financial Crisis

- We have seen that high leverage, illiquidity of assets, and liquidity of liabilities all increase the risk of trouble in the financial system. All three elements were present in 2008, creating a "perfect storm."
- The complexity of assets held by banks through a process called **securitization** was an additional undertaken by banks.
- **Mortgage-backed securities (MBS)** offered banks a mechanism by which to diversify the risk of holding an individual mortgages. MBS are interests in pools of mortgages.
- Ordinarily, a pool of mortgages has a more predictable repayment profile than an individual mortgage, because repayment of the latter is dependent on the individual circumstances of the mortgage borrower.

Complexity of CDOs

6.5 From a Housing Problem to a Financial Crisis

- In principle, the advent of MBS should have reduced the cost of mortgage borrowing by making mortgage financing more attractive to lenders.
- Securitization went further with the development of **collateralized debt obligations (CDOs)**, which became increasingly popular in the 1990s and 2000s.
- CDOs divided up the payments from a pool of mortgages into different streams—for example a senior tranche paid first, and junior tranches paid afterwards (as long as sufficient funds were available).
- The complexity of CDOs can increase substantially with further securitization. There are CDOs on CDOs and so on.

Securitization and Illiquidity of Assets

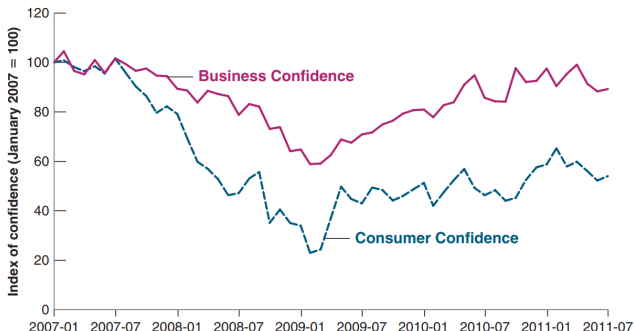
6.5 From a Housing Problem to a Financial Crisis

- Securitization created a risk that evidently was not well understood.
- The assessed value of the payment streams was contingent on housing prices continuing to rise.
- Once housing prices fell, many mortgages in a given pool were at risk, and the value of MBS and of the individual payment streams on CDOs became extremely difficult to assess.
- From a liquidity perspective, banks also relied heavily on liabilities from banks which is known as **wholesale funding**.
- Once the lending to banks through wholesale funding became a problem, many banks found themselves short of funds and were forced to sell assets.
- The securitized assets were complex and hard to value, therefore banks had to sell some assets at fire sale prices.

Macroeconomic Implications

6.5 From a Housing Problem to a Financial Crisis

- The immediate effects of the crisis were higher consumer interest rates which lowered spending and plummeting consumer confidence. The result was a sharp leftward shift in the IS curve.



Policy Responses

6.5 From a Housing Problem to a Financial Crisis

- The government implemented both fiscal and monetary policy responses to the crisis. These responses included:
 - ① The provision of higher liquidity to the financial system.
 - ② An increase in deposit insurance limits from \$100k to \$250k to prevent runs.
 - ③ Passed the **Troubled Asset Relief Program (TARP)** to clean up problem assets from banks (called the bank bailout bill).
 - ④ Implemented **unconventional monetary policy** given the fed funds rate was zero.
 - ⑤ Passed the **American Recovery and Reinvestment Act** which temporarily reduced taxes to encourage consumption and increase government spending.

The Financial Crisis and the Use of Financial, Fiscal, and Monetary Policies

6.5 From a Housing Problem to a Financial Crisis

- The net effect of these policies was to shift the IS curve back to the right and shift the LM curve down.

